you need to brush up your high-school geography before you would be able to develop location based AR apps.

## Projection Based AR

This one is more difficult to code for than location based applications. Projection based AR depends on three main factors – the surface on which projection has to be done, the light sequence to be projected and position of the projector. One point of projection based AR is that both development and implementation of an app is more expensive than a location based AR app. You are in need of an intelligent enough video editor (unless of course, you plan to project a static image), a powerful enough computer to help you with the job, a nice projector and a setup which allows you enough space to experiment. A few points which are worth a note about projection based AR:

1 You will always need a low-light situation for projection based AR to work.
2 You need enough space to develop the light sequence for projection. e.g. If you want to project your light sequence on a cube with an edge of 2 meter, kept at a distance of 8 meters from the projector, then developing the light sequence by using a dice as a reference surface is a bad idea. Ideally, the target object's size on which the simulation/testing has to be done should be the same as the target; but in case that cannot be done right away, the target object should still be large enough. It helps remove errors and allows better inspection during development.
3 Projector resolution matters. You cannot use low resolution projectors built for PowerPoint presentations in a video sequence on a concert stage.

When we talk about the domain of knowledge and experience required to undertake a projection based AR app, the skillset required is more mathematical than programmatic. Once again, it would depend on the specific application.

1 If you are planning to use projection based AR in a concert, then the skillset required is going to be almost purely creative. There is hardly any programming involved except the tool which would help generate the light sequence.
2 If you wanted to build a dialer on your palm, you would be in need of basic projection system (which would be probably mounted on your head) and a recognition system clever enough to understand what you are touching with your finger. In this case, you are more in need of image